Tutorial Sheet — TD 1

Algorithmic Complexity and Asymptotic Notation

Corresponds to: Lectures 1–2

Learning Objectives

- Define and use the **O**, **Ω** and **Θ** notations correctly.
- Classify standard growth-rate functions from slowest to fastest.
- Compute the tight complexity of simple loop-based pseudocode.

Exercise 1

Order the following functions from slowest to fastest asymptotic growth, and justify your ordering (you may use limits of ratios or known growth hierarchies):

$n!,\quad 2^n,\quad n^3,\quad n\cdot\log n,\quad \log n,\quad n^2,\quad \sqrt{n},\quad n$

Exercise 2

Give the tight bound **Θ(·)** for the number of elementary operations performed by the following pseudocode, as a function of **n**:

```
count = 0
for i = 1 to n:
    for j = i to n:
        count = count + 1
```

Exercise 3

Give the tight bound **Θ(·)** for the following pseudocode, and justify each factor of your answer separately:

```
for i = 1 to n:
    j = 1
    while j < n:
        j = j * 2
        do O(1) work
```